

Scores and Calibration Intuition

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Train a Classifier

Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Leakage

Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Locate the discrepancy for Calibration as a group property. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Calibration as a group property. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Twenty cases all predict .3 with eight positives; ten all predict .7 with seven positives. Give plot points and interpret.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Twenty cases receive score .7 and fourteen are correct. Explain what the observed rate suggests and what it does not establish about another score group.

6. Solve the given case

Ten cases receive probability .8 for spotted; eight are spotted.

Interpret calibration and its limits.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Among 20 predictions with score 0.8, 12 are correct. What is observed accuracy in this group and how does it compare with 0.8?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 14 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Add an uncertainty band to the Plant Classifier and define what the interface says when scores fall inside it.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Twenty cases receive score .7 and fourteen are correct. Explain what the observed rate suggests and what it does not establish about another score group.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Compute frequencies

Give the observed frequencies for the two worked groups.

My reasoning and evidence

13. W2 Locate the discrepancy

Which worked group overstates positive frequency and by how much?

My reasoning and evidence

14. W3 Separate ranking

Why can constant .5 predictions look calibrated but rank poorly?

My reasoning and evidence

15. W4 Apply independently

Twenty cases all predict .3 with eight positives; ten all predict .7 with seven positives. Give plot points and interpret.

My reasoning and evidence

16. W5 Protect evaluation

What is wrong with calibrating and reporting success on the same final-test records?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Scores and Calibration Intuition

Treat calibration as a comparison across groups of probability scores.

Toy classifier outputs $p(\text{spotted})=0.8$ on ten cases; eight are actually spotted. Another ten have $p=0.2$; four are spotted.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Scores and Calibration Intuition

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Group scores

Keep the ten 0.8 cases and ten 0.2 cases separate.



2. Compare frequencies

Observed spotted fractions are $8/10=0.8$ and $4/10=0.4$.



3. Interpret gap

The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

CHANGE ONE THING

The 0.2 group has two spotted cases instead of four.

Explain the changed result using the matching lesson.

Does a calibrated 0.8 score mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Treat calibration as a comparison across groups of probability scores.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

1. Group scores
2. Compare frequencies
3. Interpret gap

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Interpret gap The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. Read the picture in order. The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: The 0.2 group has two spotted cases instead of four. Which conclusion is supported by the supplied facts?

1. Its observed fraction becomes 0.2 and matches the supplied score in this group.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Its observed fraction becomes 0.2 and matches the supplied score in this group. Calibration compares stated probabilities with group outcome frequencies.

Repair a missing building block

Review Train a Classifier (ai-4/13). Goal: Connect training-only scaling and neighbour prediction in one pipeline. Short practice: Which pipeline record would help another team reproduce the result? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf predictions from training examples. Short practice: Which input distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Leakage (ai-4/8). Goal: Choose a split unit that matches the intended prediction claim. Short practice: How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
